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Kim et al.

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(54) **DISPLAY DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(52) **U.S. Cl.**
CPC **G09F 9/3026** (2013.01)
(58) **Field of Classification Search**
None
See application file for complete search history.

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(57) **ABSTRACT**

Disclosed is a display device, including a display module, a frame located in rear of the display module, a step difference adjusting unit located between the display module and the frame to adjust a distance between the display module and the frame, and a worm shaft providing a driving force to the step difference adjusting unit, the step difference adjusting unit including a holder magnet having the display module coupled to a rear surface thereof, a leveling screw extending in a rear surface direction from the holder magnet to adjust an amount of insertion into the frame, and a clutch module selectively providing the driving force of the worm shaft to the leveling screw. Accordingly, installation time may be shortened by adjusting a height or level difference between the display modules disposed adjacent to each other without separation of the display module.

14 Claims, 15 Drawing Sheets

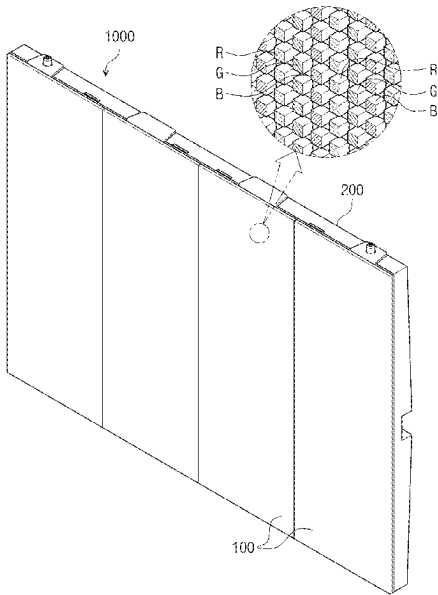


FIG. 1

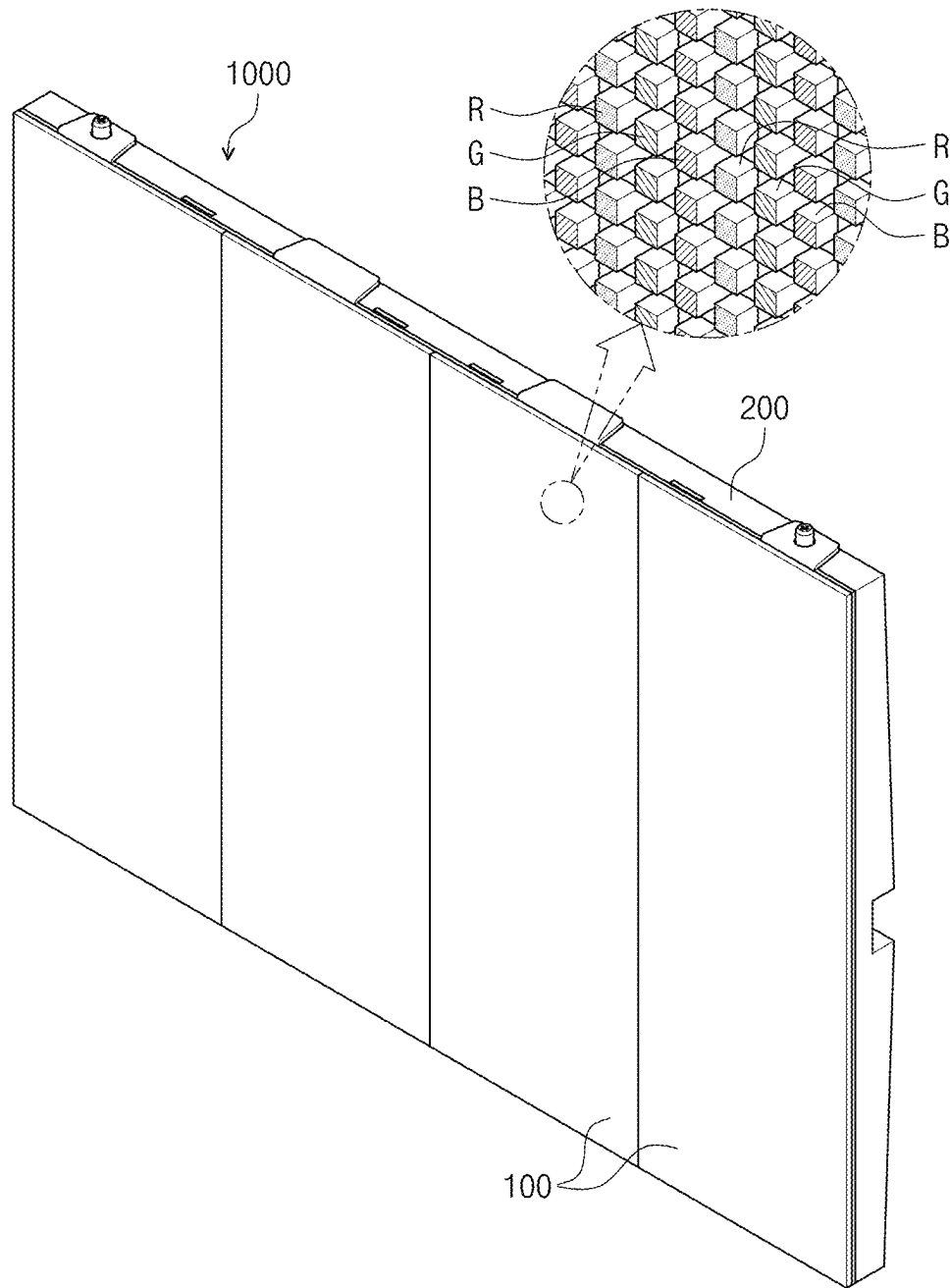


FIG. 2

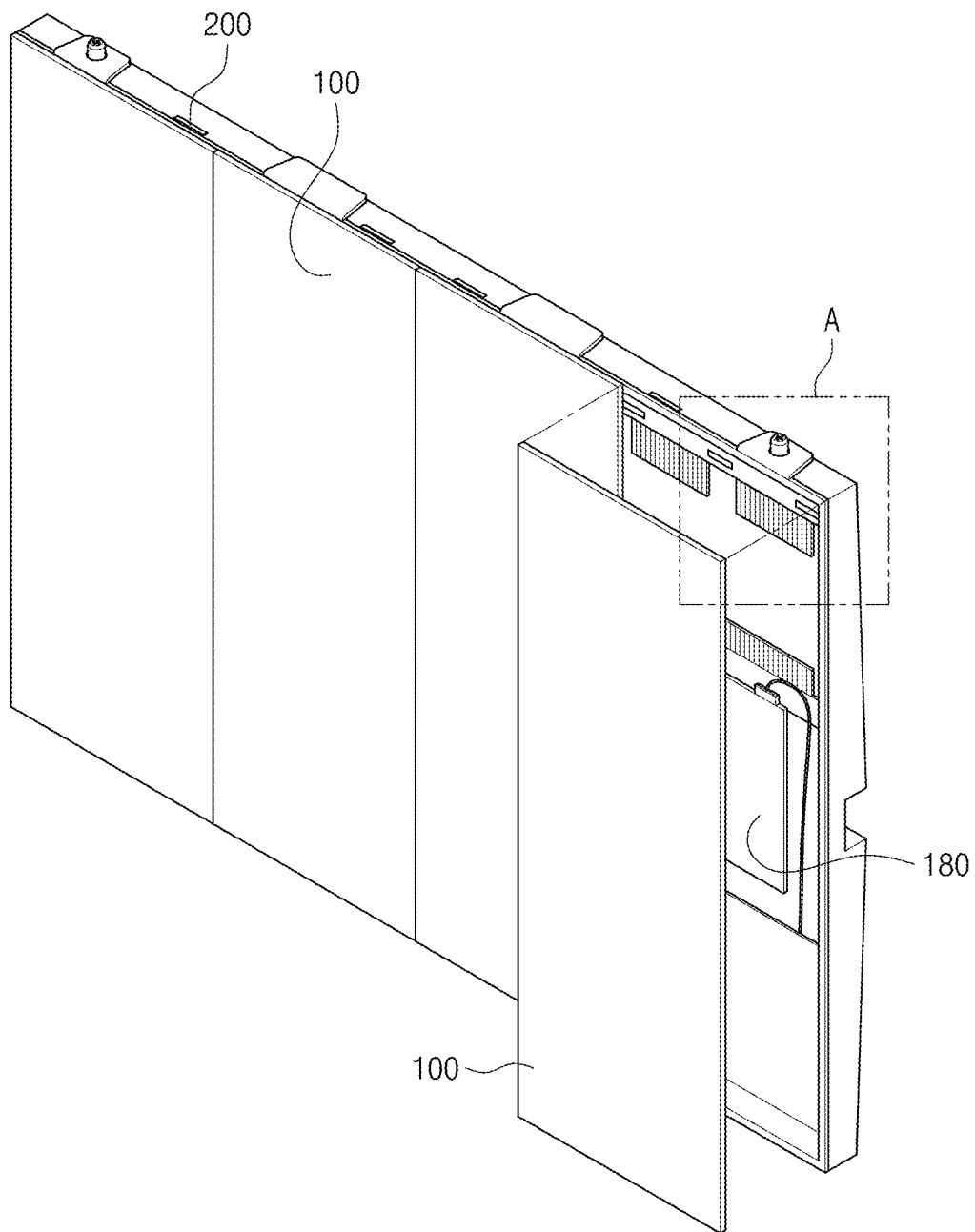


FIG. 3

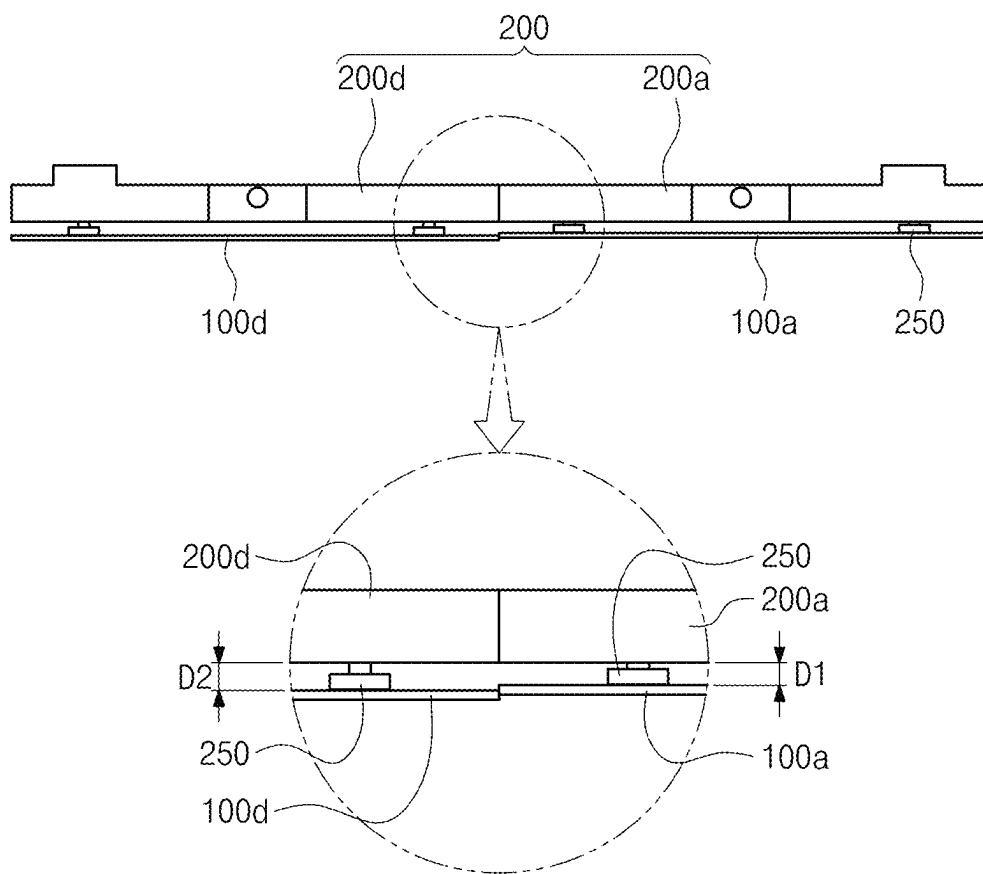


FIG. 4

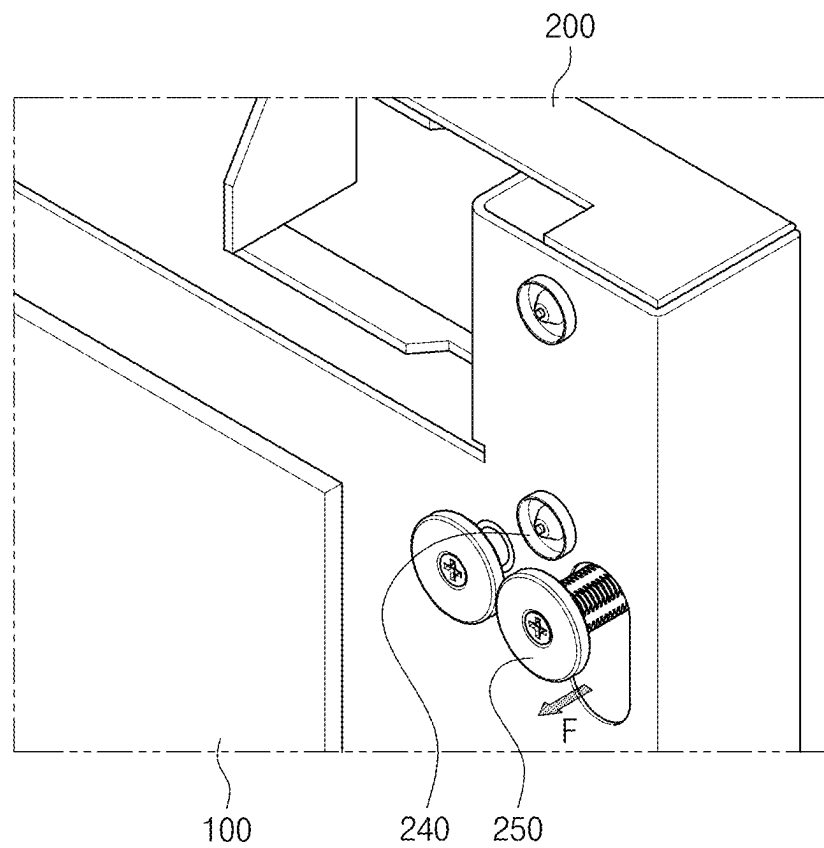


FIG. 5

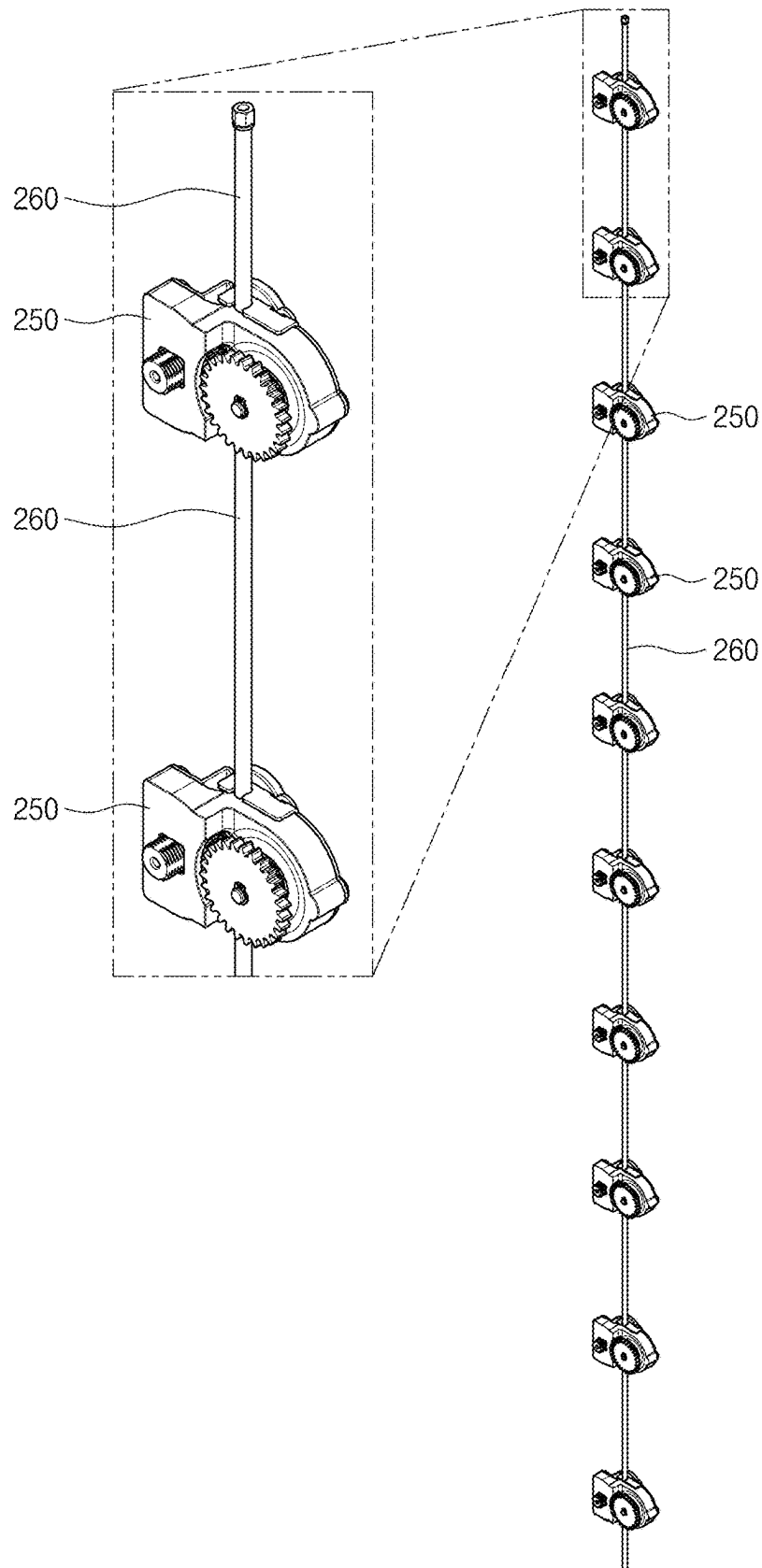


FIG. 6

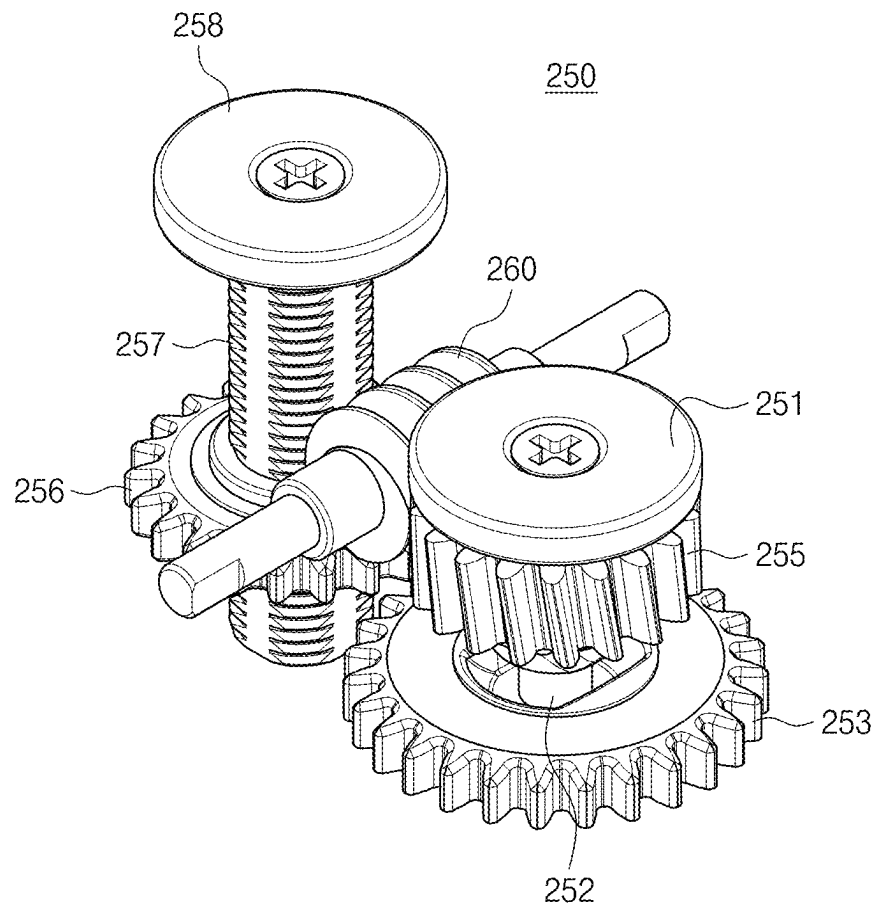


FIG. 7

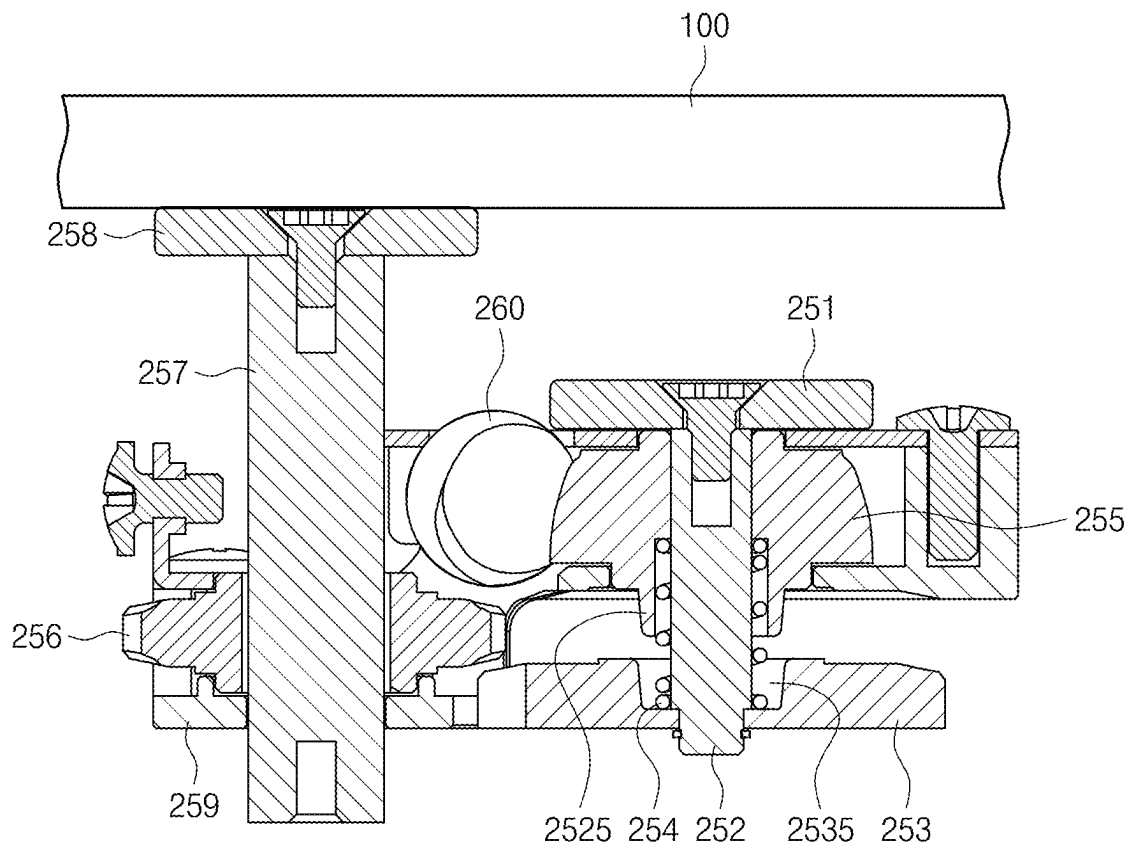


FIG. 8

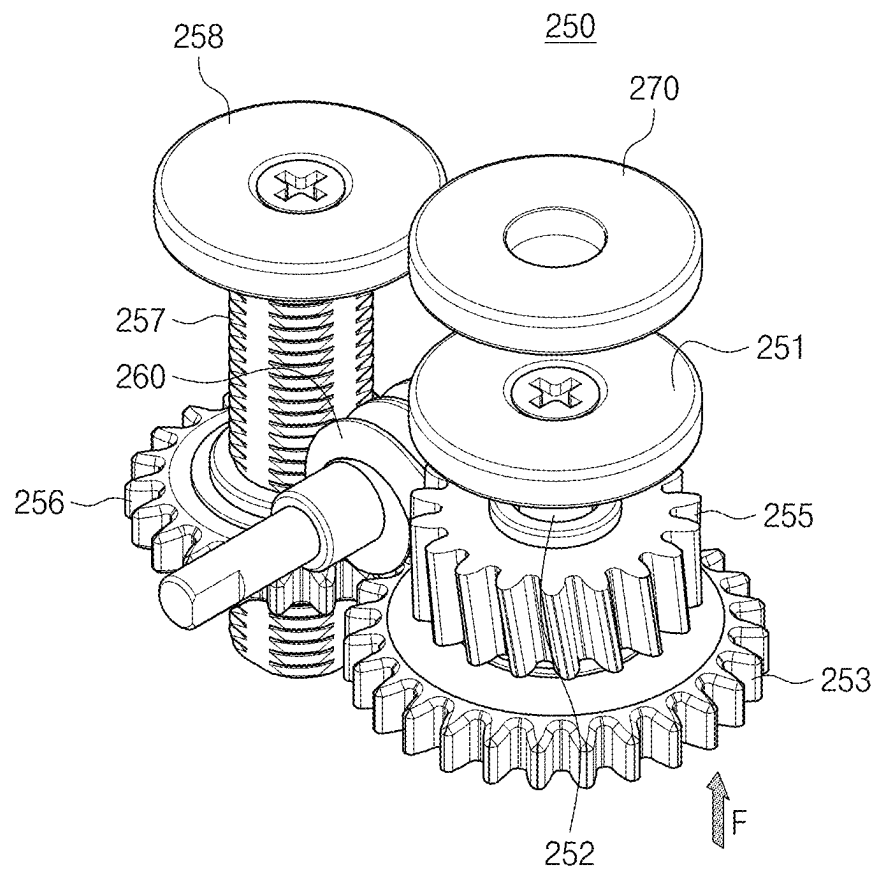


FIG. 9

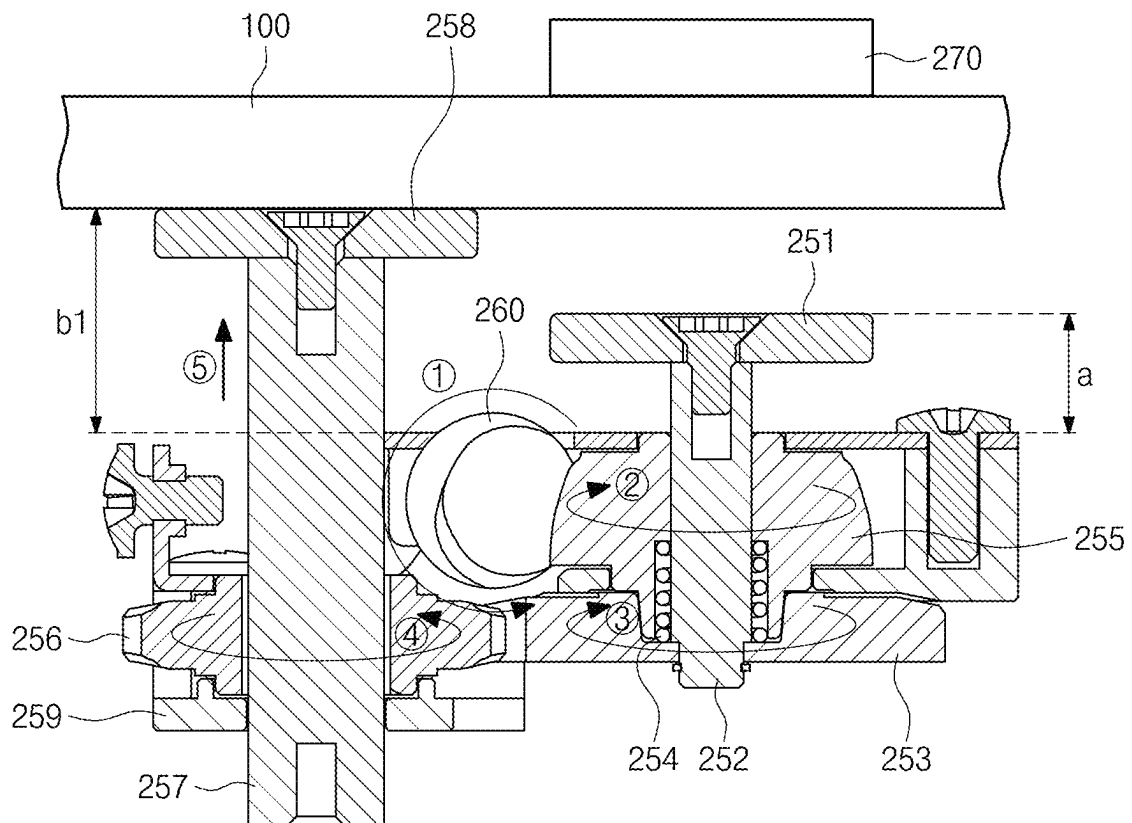


FIG. 10

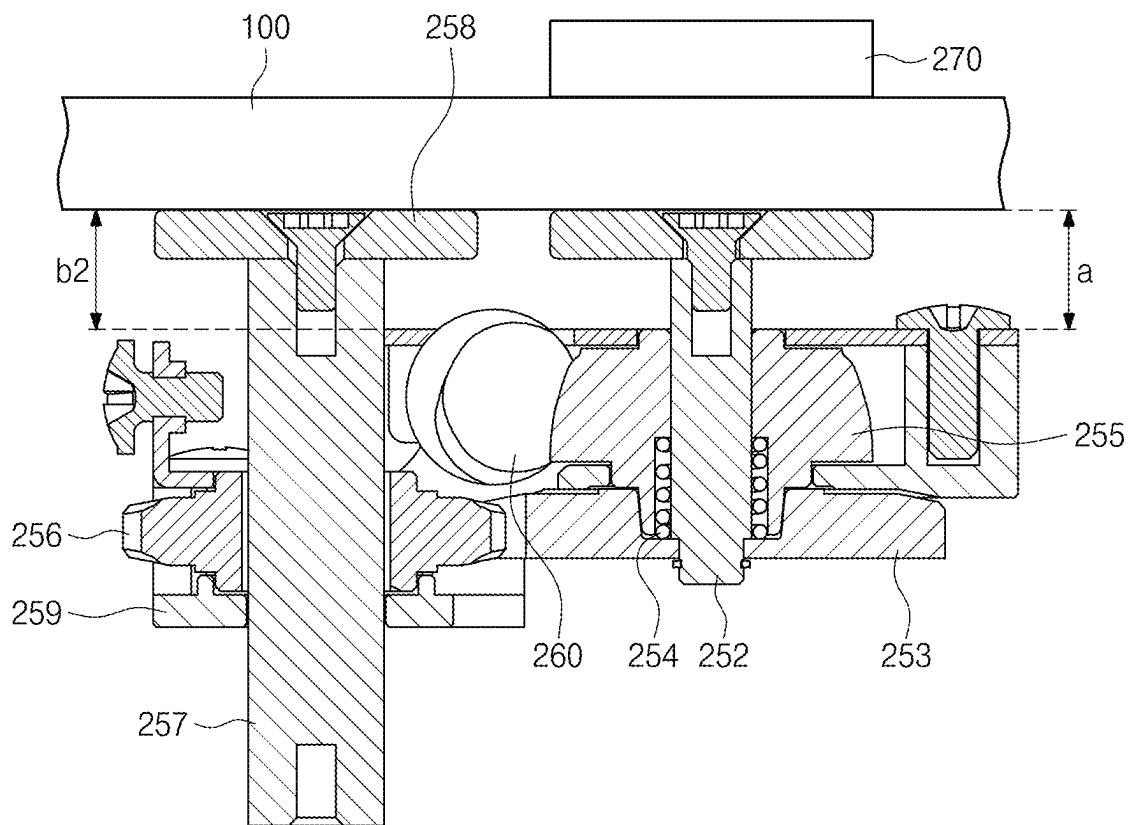


FIG. 11

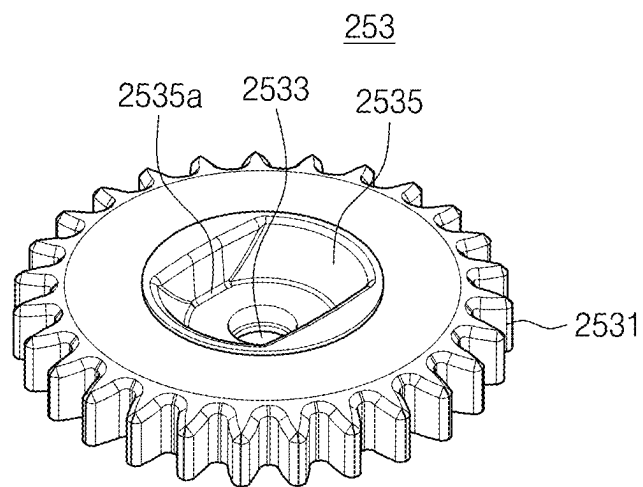


FIG. 12

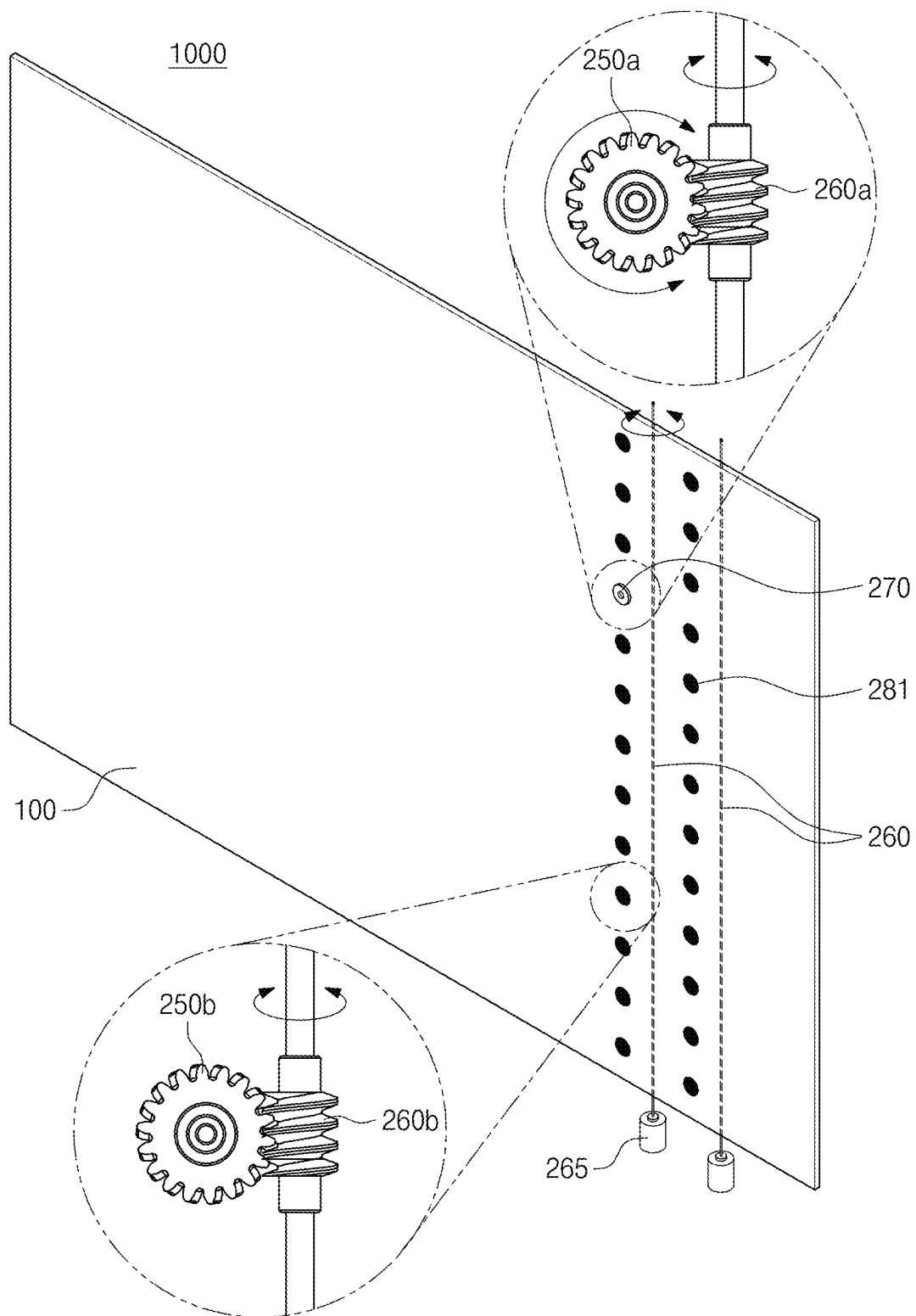


FIG. 13

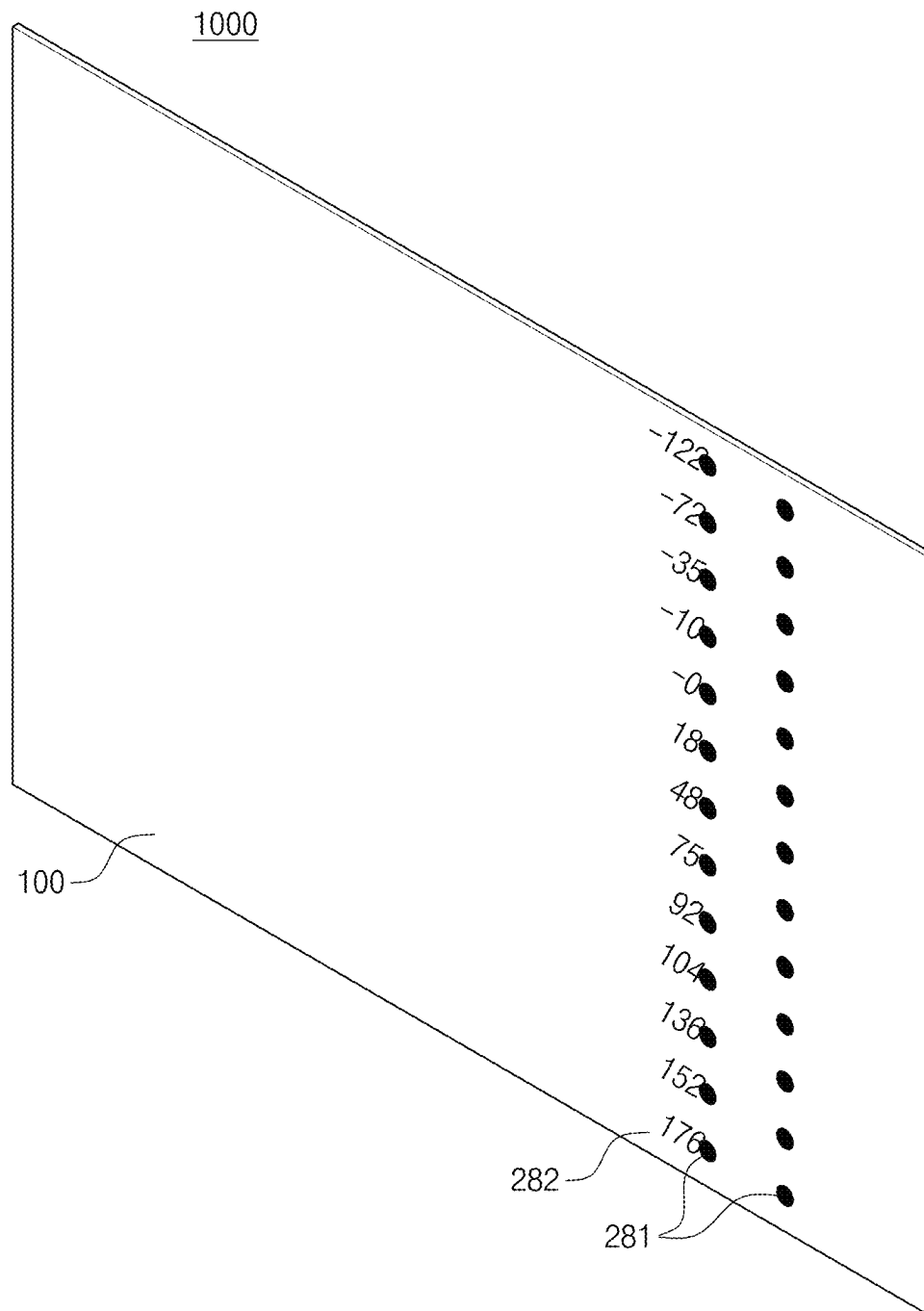


FIG. 14

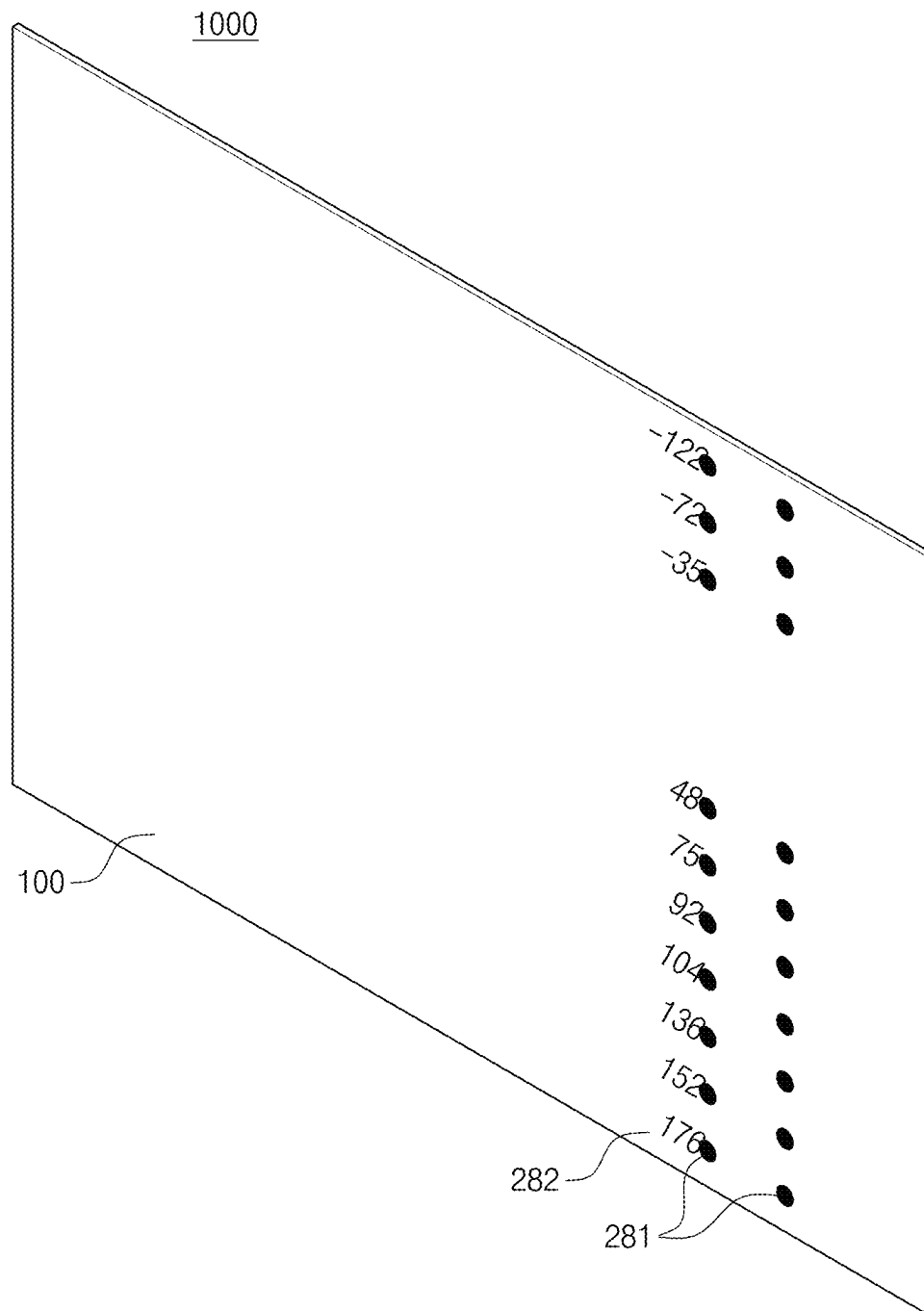
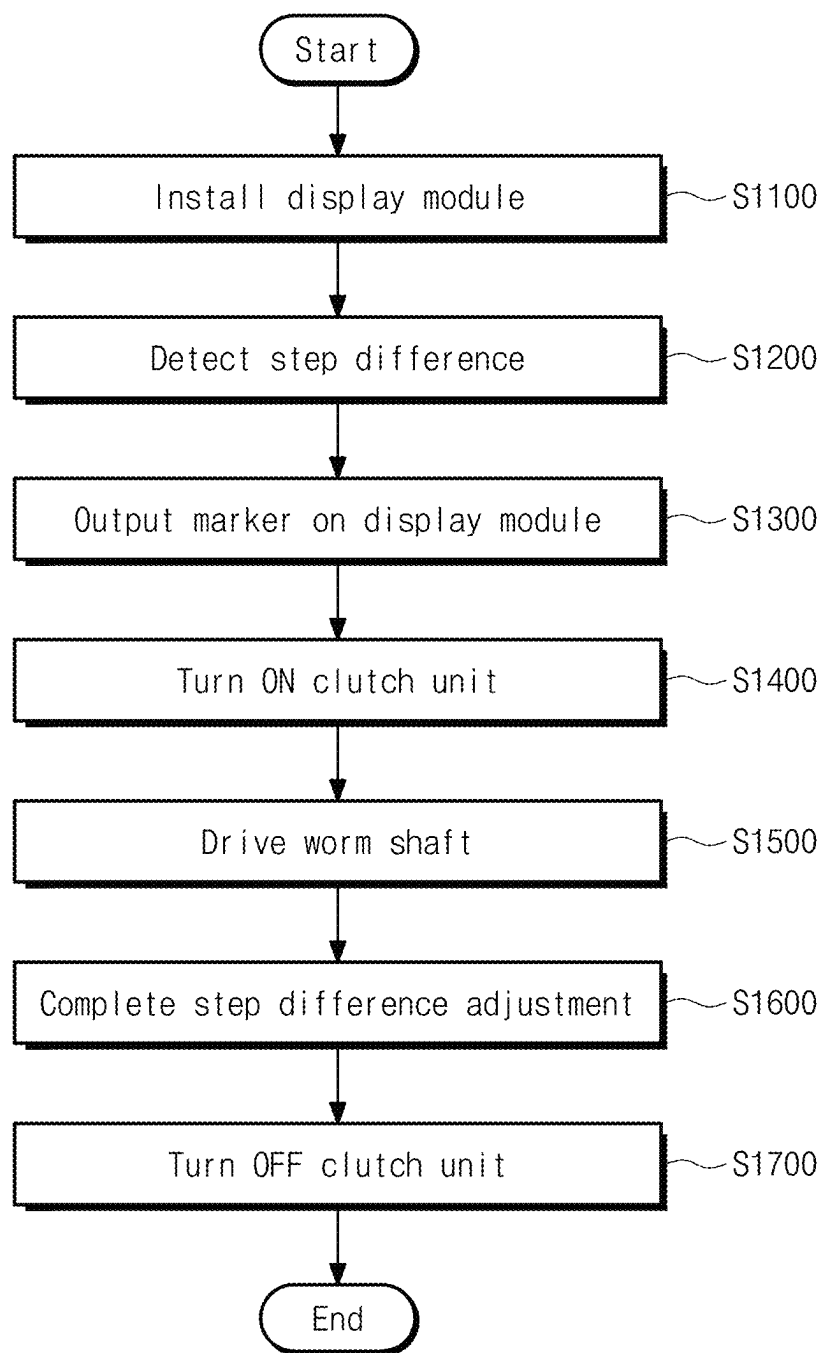


FIG. 15



1

DISPLAY DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

Pursuant to 35 U.S.C. § 119, this application claims the benefit of an earlier filing date and right of priority to International Application No. PCT/KR2024/007098, filed on May 24, 2024, the contents of which are hereby incorporated by reference herein in its entirety.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

The present embodiments relate to a display device including a plurality of display modules.

Discussion of the Related Art

Digital signage is a communication tool that can induce marketing, advertising, training effects and customer experience of companies. It is a display device that provides specific information as well as broadcast programs in public places such as airports, hotels, hospitals, and subway stations.

The digital signage is a medium that displays various content and commercial advertisements by installing display devices such as a liquid crystal display (LCD), a plasma display panel (PDP), and an organic light emitting diode (OLED) in outdoor certain places or street furniture. The digital signage can now be installed on all moving lines, where the public moves, such as apartment elevators, subway stations, subways, buses, universities, banks, convenience stores, discount stores, shopping malls.

When configuring a single screen using a plurality of display modules, it is important to configure a front position between neighboring display modules equally because there is a problem that a screen is divided or distorted due to a step difference between neighboring display modules in a vertical or horizontal direction.

Typically, when there is a step difference between neighboring display modules, the display modules are separated, a height of a coupling part is adjusted, and then the display modules are coupled together again. This method has a problem of poor accuracy because the display modules should be repeatedly coupled and separated and adjusted while the display modules are separated.

SUMMARY OF THE DISCLOSURE

One object of the present disclosure is to solve the above-mentioned problems and other problems. Another object of the present disclosure may be to provide a multi-display device capable of adjusting a height difference or a level difference between adjacent display modules.

To achieve these objects and other advantages and in accordance with the purpose of the disclosure, as embodied and broadly described herein, a display device according to the present disclosure may include a display module, a frame located in rear of the display module, a step difference adjusting unit located between the display module and the frame to adjust a distance between the display module and the frame, and a worm shaft providing a driving force to the step difference adjusting unit, the step difference adjusting unit including a holder magnet having the display module coupled to a rear surface thereof, a leveling screw extending

2

in a rear surface direction from the holder magnet to adjust an amount of insertion into the frame, and a clutch module selectively providing the driving force of the worm shaft to the leveling screw.

5 The clutch module may include a worm wheel rotating by engaging with the worm shaft and a clutch gear selectively coupled to the worm wheel to transmit the driving force of the worm wheel to the leveling screw.

10 The clutch module may include a clutch shaft passing through the clutch gear and a clutch head located on a front surface of the clutch shaft, wherein based on moving the clutch head forward, the clutch gear may be coupled to the worm wheel.

15 The display device may include a magnet jig providing a magnetic force of pulling the clutch head. Based on positioning the magnet jig on a front surface of the display module, the clutch head may be attached to a rear surface of the display module and a driving force of the worm shaft may be transmitted to the leveling screw.

20 The clutch module may further include a restoring spring applying a force in a direction of spacing the clutch gear apart from the worm wheel and the magnetic force of the magnet jig may be greater than elasticity of the restoring spring.

25 The restoring spring may be located between the clutch gear and the worm wheel by surrounding an outer circumferential surface of the clutch shaft and the worm wheel may include a spring recess for accommodating the restoring spring therein.

30 The clutch gear may be located in a rear surface direction of the worm wheel, the clutch gear may include a clutch recess formed in a rear surface of the worm wheel and a clutch protrusion protruding from a front surface of the clutch gear to be inserted into the clutch recess, and the clutch recess and the clutch protrusion may include a straight line portion.

35 The display device may further include a nut gear allowing the leveling screw to pass therethrough, and the clutch gear may move the leveling screw in front-rear direction by engaging with the nut gear in an ON state of the clutch module.

40 A plurality of the step difference adjusting units may be disposed in a longitudinal direction of the worm shaft in a manner of being spaced apart from each other and the worm shaft may be disposed to engage with a worm wheel of each of a plurality of the step difference adjusting units.

45 The display module may include a plurality of display modules and the display device may further include a displacement sensor measuring a step difference between the display modules adjacent to each other.

50 The display device may include a controller controlling the display module to output a marker at a position related to the step difference adjusting unit.

55 The controller may control the display module to output only the marker related to the step difference adjusting unit requiring step difference adjustment.

60 The controller may control the display module not to output the marker related to the step difference adjusting unit having completed step difference adjustment.

65 The display device may further include a displacement sensor measuring a step difference between the display modules adjacent to each other, and the controller may output a value measured by the displacement sensor to the display module.

According to an embodiment of the present disclosure, it is possible to adjust a height or level difference between

adjacent display modules without separation of the display modules, thereby shortening the installation time.

In addition, according to an embodiment of the present disclosure, it is possible to prevent an accident in which a display module is damaged in a process of repeating attachment and detachment.

Additional scope of applicability of the present disclosure will become apparent from the following detailed description. However, since various changes and modifications within the spirit and scope of the present disclosure can be understood by those skilled in the art, it should be understood that specific embodiments, such as the detailed description and the preferred embodiments of the present disclosure, are given as examples.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a display device according to an embodiment of the present disclosure.

FIG. 2 is a diagram illustrating a process of assembling a display module of a display device according to an embodiment of the present disclosure.

FIG. 3 is a diagram illustrating a step difference between display modules of a display device according to an embodiment of the present disclosure.

FIG. 4 is an enlarged diagram showing a portion A of FIG. 2.

FIG. 5 is a diagram illustrating a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

FIG. 6 is a perspective diagram illustrating an OFF state of a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

FIG. 7 is a cross-sectional diagram illustrating an OFF state of a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

FIG. 8 is a perspective diagram illustrating an ON state of a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

FIG. 9 is a cross-sectional diagram illustrating an ON state of a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

FIG. 10 is a cross-sectional diagram illustrating an ON state of a step difference adjusting unit of a display device according to another embodiment of the present disclosure.

FIG. 11 is a front perspective diagram illustrating a clutch gear of a display device according to an embodiment of the present disclosure.

FIG. 12 is a schematic diagram illustrating a method of controlling a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

FIG. 13 and FIG. 14 are diagrams illustrating a state in which a marker for guiding control of a step difference adjusting unit of a display device according to the present disclosure is output.

FIG. 15 is a flowchart illustrating a method of controlling a step difference adjusting unit of a display device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

Reference will now be made in detail embodiments of the present disclosure examples of which are illustrated in the accompanying drawings. Since the present disclosure may be modified in various ways and may have various forms,

specific embodiments are illustrated in the drawings and are described in detail in the present specification. However, it should be understood that the present disclosure are not limited to specific disclosed embodiments, but include all modifications, equivalents and substitutes included within the spirit and technical scope of the present disclosure.

The terms 'first', 'second', etc. may be used to describe various components, but the components are not limited by such terms. The terms are used only for the purpose of distinguishing one component from other components.

When an arbitrary component is described as "being connected to" or "being linked to" another component, this should be understood to mean that still another component(s) may exist between them, although the arbitrary component may be directly connected to, or linked to, the second component. In contrast, when an arbitrary component is described as "being directly connected to" or "being directly linked to" another component, this should be understood to mean that no component exists between them.

A singular expression can include a plural expression as long as it does not have an apparently different meaning in context.

In the present application, the terms "include" and "have" should be understood to be intended to designate that illustrated features, numbers, steps, operations, components, parts or combinations thereof exist and not to preclude the existence of one or more different features, numbers, steps, operations, components, parts or combinations thereof, or the possibility of the addition thereof.

Hereinafter, the embodiments of the present disclosure are described using a Light Emitting Diode (LED) panel as an example of a display panel. Other display panels may be used. For example, an organic light emitting diode (OLED) display panel, a plasma display panel (PDP), a field emission display (FED) panel, Liquid Crystal Display (LCD) and may be used.

FIG. 1 is a diagram illustrating a display device 1000 according to an embodiment of the present disclosure, and FIG. 2 is a diagram illustrating a process of assembling a display module of a display device 1000 according to an embodiment of the present disclosure.

Referring to FIG. 1 and FIG. 2, a multi-display device 1000 may include a display module 100 capable of displaying an image, a frame 200 supporting the display module 100, and a step difference adjusting unit 250 mounted between the display module 100 and the frame 200 to adjust a distance between them.

The display module 100 may include a display panel and a module cover positioned in rear of the display panel. The module cover may be disposed in rear of the display panel to protect a rear surface of the display panel from the outside, and may include a coupling part attached to a displacement adjusting unit to be attached to the frame.

The display panel 101 may include a plurality of pixels R, G, B. The plurality of pixels R, G, and B may be formed in each region where a plurality of data lines and a plurality of gate lines cross. The plurality of pixels R, G, B may be disposed or arranged in a matrix form.

For example, the plurality of pixels R, G, and B may include a red (R) sub-pixel, a green (G) sub-pixel, and a blue (B) sub-pixel. The plurality of pixels R, G, and B may further include a white (hereinafter, 'W') sub-pixel.

In the display panel 101, a side on which images can be observed may be referred to as a front side or a front surface. When the display panel 101 displays the images, a side on which the images cannot be observed may be referred to as a back side or a back surface. When looking at the display

5

panel **101** from the front side or the front surface, an upper part may be referred to as an upper side or an upper surface. Similarly, a lower part may be referred to as a lower side or a lower surface. Similarly, a right part may be referred to as a right side or a right surface, and a left part may be referred to as a left side or a left surface.

Regarding the display module **100**, a plurality of display modules **100** may be arranged in a manner of neighboring each other in an up-and-down direction or a left-and-right direction. The display module **100** of the present embodiment may have a long length in a vertical direction and a plurality of the display modules **100** arranged side by side in a horizontal direction.

The frame **200** may be disposed behind the plurality of display modules **100**. A front surface of the frame **200** may face the back surface of the display module **100**. The frame **200** may be disposed to correspond to the display module **100** in a thickness direction or a back-and-forth direction of the display module **100**. The frame **200** may be formed in a frame shape in which a central region is opened.

The frame **200** may be formed to be long in the up-down direction and the left-right direction so that the plurality of display modules **100** may be mounted. For example, a length of an upper side of the frame **200** may be substantially the same as a sum of a length of the upper side of the first display module **100a** and a length of the upper side of the fourth display module **100d**. A length of a right side of the frame **200** may be substantially the same as a sum of a length of the right side of the first display module **100a**, a length of the right side of the second display module **100b**, and a length of the right side of the third display module **100c**. But, it is not limited thereto. The frame **200** may be formed to be longer or shorter than the display module **100** according to an external environment such as a building or a wall to be installed.

Although FIG. **1** and FIG. **2** illustrate that a single frame **200** is formed, the present disclosure is not limited thereto. A plurality of frames **200** may be included in a size corresponding to a size of the display module.

FIG. **3** is a diagram illustrating a step difference between display modules of a display device **1000** according to an embodiment of the present disclosure. Neighboring display modules may differ from each other in thicknesses, and may have different thicknesses in a process of assembling a frame. Alternatively, a height difference on a wall for installation may occur, and thus a step difference may occur between the display modules as illustrated in FIG. **3**.

A step difference adjusting unit **250** for adjusting a distance between the display module **100** and the frame **200** is required to adjust a step difference between neighboring display modules.

The step difference adjusting unit **250** may be disposed between a plurality of display modules **100** and the frame **200**. The step difference adjusting unit **250** may be mounted on the frame **200** in a thickness direction of the display module **100**. The step difference adjusting unit **250** mounted on a front surface of the frame **200** may be attached to a rear surface of the display module **100**.

The step difference adjusting unit **250** may adjust a spaced distance between the rear surface of the display module **100** and the front surface of the frame **200**. There may be a plurality of step difference adjusting units **250**. Each of the step difference adjusting units **250** may be disposed at an end portion of the frame **200**.

The step difference adjusting unit **250** minutely adjusts an amount of protrusion from the frame **200** to adjust a distance between the frame **200** and the display module **100**. How-

6

ever, according to the related art method, since the step difference adjusting unit **250** is located on a rear surface of the display module **100**, the display module **100** must be separated from the frame **200** to control the step difference adjusting unit **250**. In the related art method, the adjustment of the step difference adjusting unit **250** is cumbersome and it is difficult to accurately grasp a step difference between neighboring display modules **100**.

Accordingly, the present disclosure provides the display device **1000** capable of adjusting the step difference adjusting unit **250** while the display module **100** is coupled to the frame **200**.

FIG. **4** is an enlarged diagram illustrating a portion A of FIG. **2**, in which the step difference adjusting unit **250** is illustrated. The step difference adjusting unit **250** of the present disclosure is located on the frame **200** and may be located on a rear surface of an end portion of the display module **100**. The step difference adjusting unit **250** may include a holder magnet **258** coupled to the display module **100**. The holder magnet **258** is projected forward from the frame **200** and is coupled to the frame **200** through a leveling screw **257**.

A projected amount of the holder magnet **258** may be adjusted by rotating the leveling screw **257** or by rotating a nut gear **256** into which the leveling screw **257** is inserted. In the related art, an amount of insertion into the frame **200** of the leveling screw **257** is adjusted by removing the display module **100** and rotating the holder magnet **258**, but in the present disclosure, the nut gear **256** into which the leveling screw **257** is inserted may be rotated using a worm shaft **260** extending in a direction vertical to the leveling screw **257** to adjust a position of the leveling screw **257** in the forward and backward directions.

FIG. **5** is a diagram illustrating the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure. The step difference adjusting unit **250** may be disposed to be spaced apart from each other by a predetermined distance along an end portion of the display module **100**.

The worm shaft **260** may be coupled to each step difference adjusting unit **250** one by one, but it is difficult for a user to directly adjust the worm shaft **260** except for an end portion of the display module **100**. Accordingly, as shown in FIG. **5**, a worm shaft **260** extended to be simultaneously coupled to a plurality of the step difference adjusting units **250** may be used.

A motor **265** (see FIG. **12**) for rotating the worm shaft **260** is provided at the end portion of the worm shaft **260** extending to the end portion of the display module **100**, or a user may manually rotate to adjust the projected amount of the leveling screw **257** of the step difference adjusting unit **250**.

However, in order to individually control only the specific step difference adjusting unit **250** by using the worm shaft **260** coupled to a plurality of the step difference adjusting units **250**, a selective synchronization structure for selectively synchronizing the worm shaft **260** with the leveling screw **257** is required. Hereinafter, the selective synchronization structure is referred to as a clutch module **251**, **252**, **253**, **254**, and **255**.

The step difference adjusting unit **250** includes a worm wheel **255** that rotates in engagement with the worm shaft **260**, and the clutch module may selectively transmit a rotational force of the worm wheel **255**, which rotates when the worm shaft **260** rotates, to the nut gear **256** into which the leveling screw **257** is inserted.

7

A state in which the clutch module transmits the rotational force of the worm wheel **255**, which rotates when the worm shaft **260** rotates, to drive the step difference adjusting unit **250** is called an ON state. A state in which the rotational force of the worm wheel **255** cannot be transmitted to the leveling screw **257** is called an OFF state because the clutch module is not synchronized with the worm shaft **260**.

FIG. **6** is a perspective diagram illustrating an OFF state of the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure, and FIG. **7** is a cross-sectional diagram illustrating an OFF state of the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure.

FIG. **8** is a perspective diagram illustrating an ON state of the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure, and FIG. **9** is a cross-sectional diagram illustrating an ON state of the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure.

Referring to FIG. **7**, a clutch module may include a worm wheel **255** rotating by being engaged with a worm shaft **260** and a clutch gear **253** transmitting a driving force of the worm wheel **255** to a leveling screw **257**.

The clutch gear **253** may be coupled to a nut gear **256** while moving in a front-rear direction. When the clutch gear **253** moves to a position engaged with the teeth of the nut gear **256**, the nut gear **256** may be rotated by receiving a rotational force of the worm wheel **255**. In this case, in order to stably transmit the rotational force of the worm wheel **255**, as shown in FIG. **9**, the clutch gear **253** may move forward and be coupled to a rear surface of the worm wheel **255**.

The clutch gear **253** moves forward and backward so as to be selectively fastened to the nut gear **256** and the worm wheel **255**. The clutch gear **253** is spaced apart from the nut gear **256** and the worm wheel **255** in the OFF state.

In the present embodiment, the worm wheel **255** and the clutch gear **253** are disposed in the front-rear direction and may be disposed adjacent to each other so as to rotate by engaging with the teeth of the worm wheel **255**. In the present embodiment, in the state in which the clutch module is in the ON state, the worm wheel **255**, the clutch gear **253**, and the nut gear **256** may be positioned at the same height in the front-rear direction and engaged with each other, thereby interoperating with each other.

A clutch shaft **252** and a clutch head **251** may be included to control the clutch gear **253** to move in the front-rear direction. A rear end of the clutch shaft **252** is coupled to the clutch gear **253**, and the clutch head **251** is positioned at a front end of the clutch shaft **252**.

The clutch head **251** includes a metal material attached to a magnet and may move forward when a magnetic force is applied from the front. When a magnet jig **270** including a magnet is positioned in front of a display module **100**, the clutch head **251** may move toward the display module **100** by a magnetic force of the magnet jig **270**.

The magnet of the magnet jig **270** may be a permanent magnet or an electromagnet that selectively generates a magnetic force when supplying power. In this case, even if the magnet jig **270** is not removed from the display module **100**, the clutch module may be switched to the OFF state simply by turning off the power of the electromagnet.

FIG. **10** is a cross-sectional diagram illustrating an ON state of the step difference adjusting unit **250** of the display device **1000** according to another embodiment of the present

8

disclosure, in which the leveling screw **257** is inserted into the nut gear **256** maximally and the leveling screw **257** is projected shortest from the frame **200**.

If a position of the clutch head **251** is projected further forward than the holder magnet **258** while the clutch gear **253** is coupled to the nut gear **256**, as shown in FIG. **10**, the clutch gear **253** may not be coupled to the nut gear **256** while the leveling screw **257** is projected short.

When switching to an ON state in which the clutch gear **253** is coupled to the nut gear **256** irrespective of a length of the leveling screw **257**, the clutch head **251** should enter an state of being in contact with the rear surface of the display module **100** as shown in FIG. **10** or not being in contact with the rear surface of the display module **100** as shown in FIG. **9**.

That is, when the clutch module is switched from the OFF state to the ON state, a moving distance of the clutch gear **253** should be shorter than a distance between the rear surface of the display module **100** and the clutch head **251**.

Accordingly, in the ON state in which the clutch gear **253** is coupled to the nut gear **256** as shown in FIG. **9** or FIG. **10**, a projected amount a from the frame **200** of the clutch shaft **252** should be equal to or less than a projected amount $b1$ or $b2$ from the frame **200** of the leveling screw **257** ($a \leq b1$ or $b2$). Hence, a moving distance a of the clutch module should be equal to or less than a minimum projected length $b2$ of the leveling screw **257** ($a \leq b2$).

In order to stably receive a rotational force of the worm wheel **255** when the clutch gear **253** is coupled to the worm wheel **255** as shown in FIG. **9** and FIG. **10**, a clutch protrusion **2525** protruding from a rear surface of the worm wheel **255** and a clutch recess **2535** formed in a front surface of the clutch gear **253** to be engaged with the clutch protrusion **2525** may be included. The clutch recess **2535** may be formed in the worm wheel **255** and the clutch protrusion **2525** may be formed at the clutch gear **253**.

After the clutch recess **2535** and the clutch protrusion **2525** are coupled, they may include a straight line portion **2535a** so that the clutch shaft **252** may rotate together when the worm wheel **255** rotates.

FIG. **11** is a front perspective view of the clutch gear **253** of the display device **1000** according to one embodiment of the present disclosure, and the clutch recess **2535** is illustrated. The clutch recess **2535** may have a polygonal shape or a shape formed to have a straight line partially in a circular shape.

A shape of the clutch recess **2535** is configured to be symmetrical to facilitate the worm wheel **253** to be coupled to the clutch gear **252** irrespective of the disposition of the worm wheel **255**. The clutch recess **2535** of FIG. **11** is symmetrical to a 180° section, but the straight line portion **2535a** may be disposed as a 90° section or a 60° section.

Since the clutch shaft **252** is fastened to the clutch recess **2535**, a shaft hole **2533** to which the clutch shaft **252** is coupled may be formed in the clutch recess **2535**. The clutch gear **253** is moved forward when the clutch head **251** is moved forward by the magnet jig **270**, and the clutch protrusion **2525** of the worm wheel **255** is inserted into the clutch recess **2535** of the clutch gear **253**. By receiving the rotational force of the worm wheel **255**, the clutch gear **253** may be rotated, and the nut gear **256** may be rotated to move the leveling screw **257** in the front-rear direction.

The clutch shaft **252** may serve as a rotating shaft of the worm wheel **255** and the clutch gear **253**, and may be fixed to both the clutch shaft **252** and the worm wheel **255** to rotate together, or may move only forward and backward without rotating.

When the clutch shaft **252** is fixed to both the worm wheel **255** and the clutch gear **253**, the clutch shaft **252** may transmit the rotational force of the worm wheel **255** to the clutch gear when the worm wheel **255** is rotated. In this case, when the worm wheel **255** is rotated in the OFF state of the clutch module, both the clutch gear **253** and the clutch shaft **252** are rotated. Since the clutch gear **253** is not fastened to the nut gear **256**, the leveling screw **257** is not moved.

However, in this case, since the driving force of the worm shaft **260** is unnecessarily consumed, the worm wheel **255** and the clutch shaft **252** may be configured not to rotate together but only the worm wheel **255** may be configured to rotate around the clutch shaft **252**. Since the worm shaft **260** is not fixed to the worm wheel **255**, the rotational force of the worm wheel **255** is not transmitted to other components in the OFF state.

As shown in FIG. 8, when the clutch gear **253** is fastened to the worm wheel **255**, the teeth **2531** of the clutch gear **253** are engaged with the teeth of the nut gear **256** coupled with the leveling screw **257**. Therefore, the rotation of the worm wheel **255** is transmitted to the nut gear **256** through the clutch gear **253**.

After the display module **100** is moved by a desired amount, the magnet jig **270** may be spaced apart from a front surface of the display module **100** to stop the movement of the leveling screw **257**.

When the magnet jig **270** is spaced apart from the display module **100**, the clutch gear **253** of the clutch module is spaced apart from the worm wheel **255** and the clutch module is switched to the OFF state. In this case, a restoring spring **254** for pressing the clutch gear **253** in a rear direction may be further included to provide a force for the clutch gear **253** to be moved backward.

In the present embodiment, the restoring spring **254** may be positioned between the worm wheel **255** and the clutch gear **253**, may be compressed in a state where the clutch gear **253** and the worm wheel **255** are coupled together as shown in FIG. 9, and may be tensioned in a state where they are spaced apart as shown in FIG. 7.

A magnetic force of the magnet jig **270** may provide a magnetic force greater than the elasticity of the restoring spring **254**, so that the magnet jig **270** may move the clutch head **251** forward to compress the restoring spring **254**.

The restoring spring **254** may not be coupled to at least one of the worm wheel **255** or the clutch gear **253**. When another step difference adjusting unit **250** is driven, the worm wheel **255** of the step difference adjusting unit **250** in the OFF state rotates, and thus this is to prevent the rotational force of the worm wheel **255** from being transmitted to the clutch gear **253** through the restoring spring **254**.

FIG. 12 is a schematic diagram illustrating a method of controlling the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure. As described above, the worm shaft **260** may be elongated to control a plurality of the step difference adjusting units **250**, and a plurality of the step difference adjusting units **250** may be disposed along a longitudinal direction of the worm shaft **260**.

As only the step difference adjusting unit **250a** that turned on the clutch module with the magnet jig **270** is synchronized with the worm shaft **260**, the clutch gear **253** rotates to drive the leveling screw **257**, and the rest of the step difference adjusting units **250b** are not driven even if the worm shaft **260** rotates.

The display device **100** of the present disclosure may control the step difference adjusting unit **250** without sepa-

rating the display module **100**, but there is a problem that it is difficult for a user to find a position of the clutch head **251** of the step difference adjusting unit **250** in front of the display module **100**.

Accordingly, the display device **1000** of the present disclosure outputs a marker **281** displaying the position of the clutch head **251** of the step difference adjusting unit **250** on a front surface of the display module **100** so that a user may easily recognize a contact position of the magnet jig **270**.

A user may visually check a step difference between the neighboring display modules **100**, contact the magnet jig **270** with the marker **281** corresponding to the step difference adjusting unit **250** where the step difference needs to be adjusted, and then rotate the worm shaft **260** to adjust the step difference.

The marker **281** may be displayed on a screen in white or color. When the display module **100** is brightly turned on for aging after installing the display module **100** on the frame **200**, the marker **281** may be displayed in dark color on the screen of the display module **100**.

Alternatively, the display device **1000** may further include a displacement sensor **240** that measures a step difference between neighboring display modules **100**. As shown in FIG. 4, the displacement sensor **240** may measure a step difference between the display modules **100** by embedding an optical sensor such as an ultrasonic wave or a laser in a rear surface of the display module **100** or the frame **200**.

Alternatively, since the displacement sensor **240** is no longer used after it is installed, a jig-shaped displacement sensor (not shown) that is installed in the front surface of the display module **100** may be used only when the display module **100** is installed in the frame **200**.

FIG. 13 and FIG. 14 are diagrams illustrating a state in which the marker **281** for guiding control of the step difference adjusting unit **250** of the display device **1000** according to the present disclosure is outputted.

As shown in FIG. 13, a controller **180** may output a value **282** measured by the displacement sensor **240** to the display module **100** and output the step difference adjusting unit **250** to be adjusted. The displacement sensor **240** may detect not only a step difference between the neighboring display modules **100** but also an inclination of the display module **100** and guide a single display module **100** to be installed while forming a plane.

All markers **281** are not always displayed. As shown in FIG. 14, the marker **281** corresponding to the adjustment-completed step difference adjusting unit **250** may be no longer outputted, or the marker **281** corresponding to the step adjustment unit **250** having the same height as the neighboring display module **100** and requiring no adjustment may not be outputted.

FIG. 15 is a flowchart illustrating a method of controlling the step difference adjusting unit **250** of the display device **1000** according to an embodiment of the present disclosure. After the display module **100** is installed in the frame **200** (**S1100**), a step difference from a neighboring display module **100** may be sensed by using naked eyes or the displacement sensor **240** (**S1200**).

The controller may output the marker **281** indicating a position of the clutch head **251** of the step difference adjusting unit **250** on the display module **100** (**S1300**, see FIG. 13). In this case, an adjustment-required step difference detected by the displacement sensor **240** may be displayed numerically on the display module **100**.

If the magnet jig **270** contacts the marker **281** at a step difference adjustment required position, the clutch unit of

11

the corresponding step difference adjusting unit **250** is switched to the ON state (**S1400**, see FIG. **8**). When the worm shaft **260** is driven (**S1500**), only the leveling screw **257** of the step difference adjusting unit **250** having the clutch unit switched to the ON state may be moved to adjust the step difference of the display module **100**. When the step difference adjustment is completed by rotating the worm shaft **260** by a required movement amount (**S1600**), the clutch unit may be switched to the OFF state by spacing the magnet jig **270** apart from the display module **100** (**S1700**).

When it is necessary to adjust a plurality of the step difference adjusting units **250**, the magnet jig **270** may be moved to another marker **281** and the steps **S1400** to **S1700** may be performed repeatedly.

As discussed above, according to an embodiment of the present disclosure, the installation time may be shortened by adjusting the height difference or the level difference between the display modules **100** disposed adjacent to each other without separation of the display module **100**.

In addition, according to an embodiment of the present disclosure, it is possible to prevent an accident in which the display module **100** is damaged in a process of repeating attachment and detachment.

The above detailed description needs not be construed to be restrictive in any aspect and needs to be considered illustrative. The scope of the present disclosure needs to be determined by reasonable interpretation of the appended claims, and all changes within the equivalent scope of the present disclosure are included in the scope of the present disclosure.

What is claimed is:

1. A display device, comprising:

a display;

a frame located in rear of the display;

a step difference adjusting device located between the display and the frame and configured to adjust a distance between the display and the frame; and

a worm shaft configured to provide a driving force to the step difference adjusting device, the step difference adjusting device comprising:
a holder magnet having the display coupled to a rear surface thereof;

a leveling screw extending in a rear surface direction from the holder magnet, the leveling screw configured to adjust an amount of insertion into the frame; and

a clutch configured to selectively provide the driving force of the worm shaft to the leveling screw.

2. The display device of claim 1, the clutch comprising:
a worm wheel configured to rotate by engaging with the worm shaft; and

a clutch gear selectively coupled to the worm wheel and configured to transmit the driving force of the worm wheel to the leveling screw.

3. The display device of claim 2, the clutch comprising:
a clutch shaft passing through the clutch gear; and
a clutch head located on a front surface of the clutch shaft, wherein based on moving the clutch head forward, the clutch gear is coupled to the worm wheel.

12

4. The display device of claim 3, comprising a magnet jig configured to provide a magnetic force of pulling the clutch head, wherein based on positioning the magnet jig on a front surface of the display, the clutch head is attached to a rear surface of the display and a driving force of the worm shaft is transmitted to the leveling screw.

5. The display device of claim 4, wherein the clutch further comprises a restoring spring configured to apply a force in a direction of spacing the clutch gear apart from the worm wheel, and wherein the magnetic force of the magnet jig is greater than elasticity of the restoring spring.

6. The display device of claim 5, wherein the restoring spring is located between the clutch gear and the worm wheel and surrounds an outer circumferential surface of the clutch shaft, and wherein the worm wheel comprises a spring recess that accommodates the restoring spring therein.

7. The display device of claim 2, wherein the clutch gear is located in a rear surface direction of the worm wheel, wherein the clutch gear comprises a clutch recess formed in a rear surface of the worm wheel and a clutch protrusion protruding from a front surface of the clutch gear so as to be inserted into the clutch recess, and wherein the clutch recess and the clutch protrusion comprise a straight line portion.

8. The display device of claim 2, further comprising a nut gear arranged to allow the leveling screw to pass there-through, wherein the clutch gear is configured to move the leveling screw in front-rear direction by engaging with the nut gear in an ON state of the clutch.

9. The display device of claim 1, wherein the step difference adjusting device is one of a plurality of the step difference adjusting devices that are disposed in a longitudinal direction of the worm shaft in a manner of being spaced apart from each other, and wherein the worm shaft is disposed to engage with a worm wheel of each of the plurality of the step difference adjusting devices.

10. The display device of claim 1, wherein the display comprises a plurality of displays, and wherein the display device further comprises a displacement sensor measuring a step difference between displays of the plurality of displays that are adjacent to each other.

11. The display device of claim 1, further comprising a controller configured to control the display to output a marker at a position related to the step difference adjusting device.

12. The display device of claim 11, wherein the marker that is output by the display is a marker that is related to the step difference adjusting device when the step difference adjusting device requires a step difference adjustment.

13. The display device of claim 11, wherein the controller controls the display not to output the marker related to the step difference adjusting device having completed step difference adjustment.

14. The display device of claim 11, further comprising a displacement sensor measuring a step difference between the displays adjacent to each other, wherein the controller outputs a value measured by the displacement sensor to the display.

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